COVID_19 STATE-LEVEL DATA VISUALIZATIONS
(REPORT)

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MASTERS PROGRAM

GLOBAL PUBLIC HEALTH

SUBJECT: DIGITAL HEALTH

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LINKS:

GITHUB REPOSITORY

SHINY APPLICATION

Introduction:

During COVID-19, large amounts of health, economic, demographic, and social data were collected. Need to convert that data into presentable forms arose to use it for public health decision makings. Interactive dashboards became popular which were produced after cleaning, analyzing, and exploring those data. Such visualizations provide detailed explanations and comparison of information in demand.

Context:

I have tried to create an interactive shiny application using COVID_19 data of United States of America available on the Kaggle website. This application is built considering epidemiological factors (infections and deaths), health system capacity (ICU beds), social factors (income, population), and environmental factors (pollution). The purpose was to explore the dependency of one factor on another. Such as: Relation of infection with number of deaths, ICU capacity in one state as compared to the other, pollutions levels in different states. These visualizations in the form of graphs and charts will provide these comparisons and impacts across the region of America.

Data used is taken from the CSV file called (COVID19_state.cvs), which is also called raw data. It includes State name, Number of COVID-19 tests, infections, deaths, Population size, Income, ICU beds, Pollution levels, and other indicators.

Packages used in R studio for the project are:

- Pacman __ for package management
- shiny __ for building the interactive web application
- tidyverse __ for data manipulation and visualization
- readr __ for reading CSV files
- Dplyr __ for data transformation
- ggplot2 __ for creating plots
- janitor __ for cleaning variable names
- plotly __ for interactive visualizations

Data Cleaning and Preparation:

The raw was cleaned and analyzed before working on it. `Clean_names()` function was used to convert column names and remove spaces, to make data easier to work on R studio. Two indicators were derived using `mutate()` function:

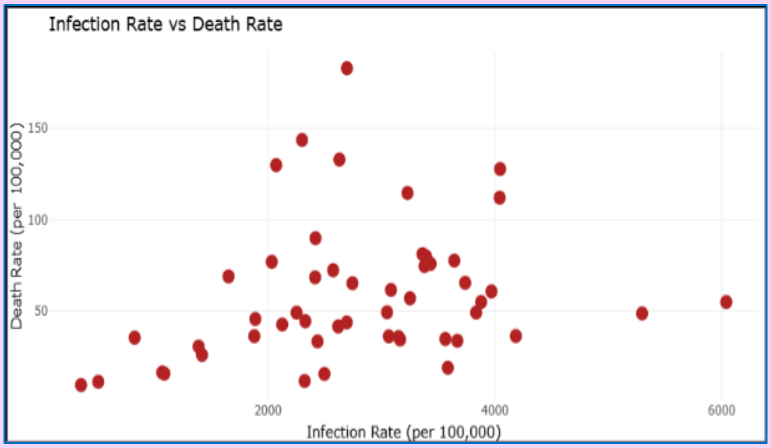
- **Infection rate**: number of infected cases per 100,000 population.
- **Death rate**: number of deaths per 100,000 population.

They made raw data meaningful and allowed easier comparison across the states. Limited number of variables were used to work on. `Drop_na()` function was used to remove missing values to make sure that visualizations are based on complete data. All these functions were used to make data set clear and well-organized, which was then called `covid_clean.`

Visualization Design:

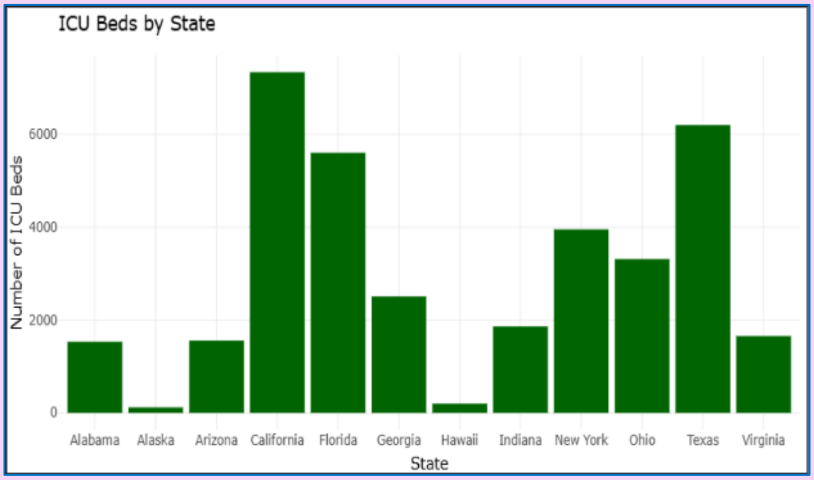
Tab 1: Infection vs Death (Scatter Plot)

This tab displays a scatter plot of infection vs death rate for the states. The **purpose** is to explore the widespreadness of COVID-19 infections and their mortality rates. States can be filtered based on **income range** and **population range** using slider inputs. Infection rate is on the x-axis and death rate is on the y-axis. We can hover over points and explore the data. This visualization helps to assess whether higher infection rates are associated with higher death rates and how this relationship changes across income and population levels.



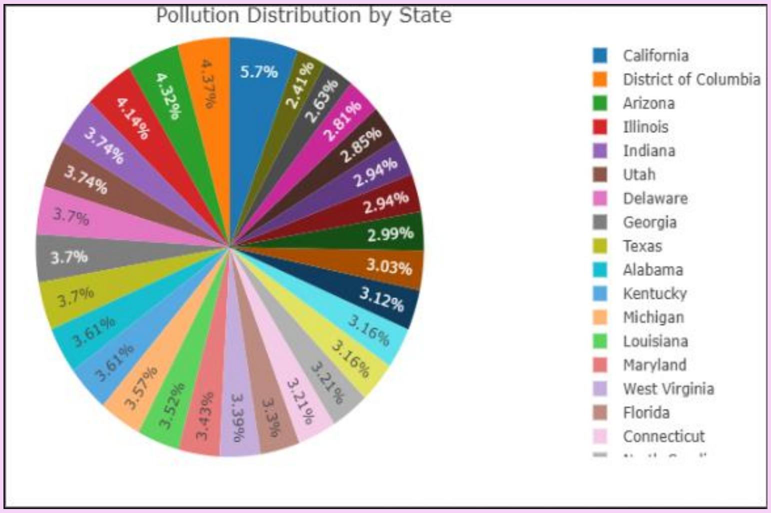
Tab 2: ICU Capacity (Bar Chart)

This tab shows a bar chart of ICU beds by state. The **purpose** is to compare ICU beds across states to help determine disparities in healthcare infrastructure. Specific states and their ICU bed capacity can be selected to determine the range. The resulting bar chart shows the number of ICU beds for the selected states. Hover over the bars for exact values. This shows differences in healthcare capacity in the states selected in filter area.



Tab 3: Pollution Pie Chart

This tab presents a pie chart showing the distribution of pollution levels in selected states. The **purpose** is to show that environmental factors such as pollution are potential contributors to respiratory issues, so, through this graph, pollution levels across regions can be visually compared. States and a pollution range can be filtered. Pollution values and percentage contribution of each state to total pollution among the selected group can be seen on the pie chart.



Conclusion:

The project shows that an interactive Shiny dashboard can be used to explore COVID-19 data in multi-dimensional ways by combining health system capacity and environmental indicators instead of just focusing on case counts.